

Analysis Memo

Purpose of the Analysis

The purpose of this analysis is to evaluate how prompt design and model choice affect the quality of language model outputs for research-oriented tasks. In Phase 1, I focused on two tasks: Claim–Evidence Extraction and Cross-source Synthesis. These tasks were chosen because they are common in academic research and require careful grounding in source material. By comparing a baseline prompt (Prompt A) with an improved, structured prompt (Prompt B) across two models, this analysis aims to understand how prompt constraints influence grounding, evidence use, and overall reliability.

Impact of Prompt Design

The most important finding from Phase 1 is that prompt design has a strong impact on output quality. Prompt A, which provided minimal instructions, often produced fluent and readable responses, but these outputs were usually high-level and difficult to verify. In contrast, Prompt B required a fixed output structure, explicit citations, and allowed the model to state when evidence was insufficient. This led to outputs that were more grounded, more transparent, and easier to evaluate. Across all test cases, Prompt B consistently reduced unsupported claims and improved the connection between conclusions and evidence.

Task 1: Claim–Evidence Extraction

In Task 1, Prompt A was able to identify relevant claims from both academic papers and industry reports, but the supporting evidence was usually paraphrased rather than directly quoted. This made it hard to confirm whether the evidence truly supported each claim. Prompt B significantly improved performance by forcing the model to extract verbatim evidence snippets and attach chunk-based citations. As a result, the outputs were easier to verify and could be reused directly for analysis. This comparison shows that explicit evidence requirements are critical for research tasks that depend on source accuracy.

Task 2: Cross-source Synthesis

For Task 2, Prompt A produced summaries that correctly captured general differences between sources, but often merged conclusions without clearly indicating which source supported which point. This resulted in overgeneralized synthesis and weak traceability. Prompt B addressed this problem by separating agreement and disagreement into a structured table and requiring explicit source attribution. The improved prompt made differences between academic studies and industry case studies much clearer and prevented the model from combining claims without evidence. This demonstrates that structured synthesis prompts are essential when comparing multiple sources.

Failure Modes Observed

Several common failure modes were observed during Phase 1. With Prompt A, the most frequent issues were vague paraphrasing, overgeneralized synthesis, and missing evidence attribution. These problems occurred across both tasks and both models. Prompt B greatly reduced these failures by introducing explicit constraints, such as fixed output formats, citation rules, and the option to state “Insufficient evidence.” This suggests that many model errors are not due to lack of capability, but to insufficient guidance in the prompt.

Model Comparison

Both ChatGPT 5.2 and Claude Haiku 4.5 showed similar behavior patterns throughout the evaluation. When using Prompt A, both models produced high-level but weakly grounded outputs. When using Prompt B, both models generated structured, well-grounded, and verifiable responses. There was no major difference in failure types between the two models. This indicates that prompt design had a greater influence on output quality than model choice in this phase.

Implications for Phase 2

The findings from Phase 1 provide clear guidance for Phase 2 system design. Structured prompts with explicit citation requirements should be used by default. The system should also allow the model to acknowledge missing evidence rather than forcing a complete answer. In addition, retrieval mechanisms should be integrated to control which sources the model can access, further improving grounding and reliability. These design choices will help ensure that the Personal Research Portal supports evidence-based reasoning rather than surface-level summarization.

Conclusion

In conclusion, Phase 1 shows that careful prompt design is essential for using language models in research settings. While baseline prompts can generate readable summaries, they are not sufficient for tasks that require strong grounding and verifiable evidence. Improved prompts with clear structure and constraints consistently produce more reliable and useful outputs. These insights form a solid foundation for building the Personal Research Portal in later phases.